Reading Environment and History through Tree Rings

Dendroclimatology

Lesson 7, 90 minutes. Testing climate evidence

Learning objectives

Explain why climate before instrumental records matters.

Relate species, site and season to calibration and verification.

Explain the scope and limits of a reconstruction.

NOAA: paleoclimate data

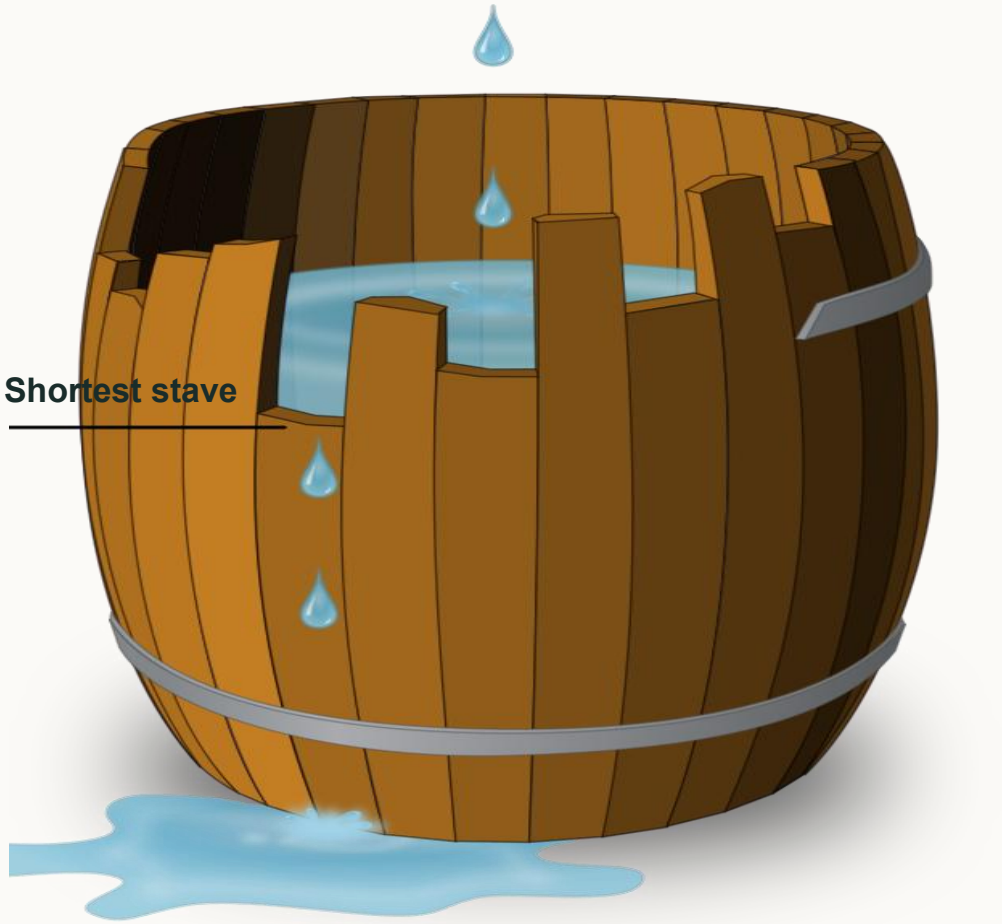
Why reconstruct past climate?

Place recent conditions in a longer history.

Investigate extreme years by region and season.

Examine climate mechanisms and model performance.

Limiting factors: Liebig’s barrel



The shortest stave limits the water level.

The strongest constraint can vary by place and season.

Temperature, water and other factors interact and may jointly limit growth.

The barrel is a conceptual analogy; stave heights are not measurements.

Image: DooFi / Wikimedia Commons (public domain)

Tree rings and limiting factors: Fritts / LTRR

Study: Stine & Huybers (2019)

Choosing species and sites

Choose species and sites sensitive to the target climate.

Record age, slope, soils, crown position and history.

Replicate among trees and across nearby sites.

R52 Yonenobu & Eckstein (2006)

Previous-year effects and seasonality

Stored resources, foliage and buds can affect next year's growth.

Previous summer and current spring can act through different pathways.

Consider biological seasons as well as annual means.

R52 Yonenobu & Eckstein (2006)

Which tree-ring properties to measure

Ring width measures radial wood increment.

Density and anatomy can record different processes.

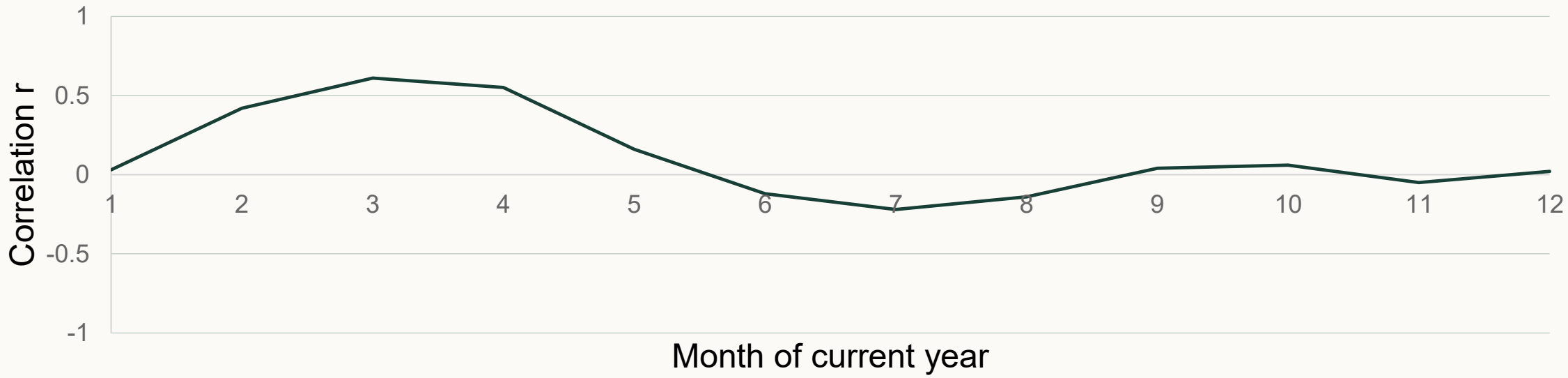
Stable isotopes reflect physiology and water as well as climate.

R52 Yonenobu & Eckstein (2006)

Reading monthly climate responses

A fictional example of current-year monthly correlations.

Read the sign and seasonal pattern.



Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Season selection and hypotheses

Define the season, region and climate variable.

Searching many months or models can select chance correlations.

Evaluate the selection procedure using independent periods.

Checking instrumental climate data

Check distance, elevation and terrain between station and forest.

Read metadata on missing values and observational changes.

Distinguish observations, interpolated values and anomalies.

R52 Yonenobu & Eckstein (2006)

Chronology signal and sample depth

Check dating and measurement quality.

Multiple cores from one tree are not independent trees.

Show annual sample depth and stability of the shared signal.

R52 Yonenobu & Eckstein (2006)

Detrending and retained variation

Choose how to handle age- and size-related trends.

Flexible detrending can remove long climate variations.

Test whether conclusions change under alternative processing.

R52 Yonenobu & Eckstein (2006)

Chronology length and long timescales

A long chronology may comprise many shorter individual series.

Series length and detrending constrain recoverable slow variation.

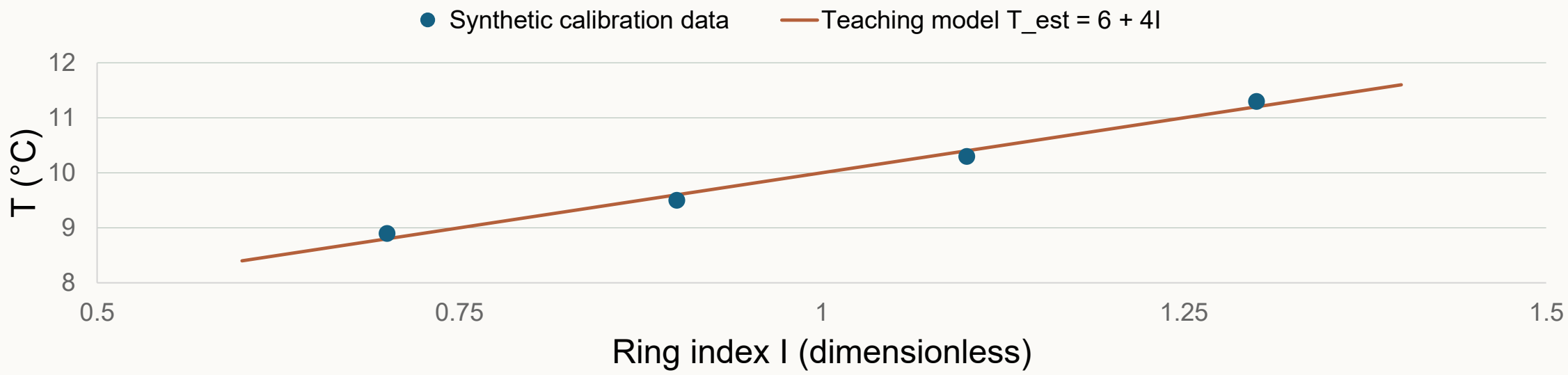
This is discussed as the segment length curse.

R54 Cook et al. (1995)

Regression for calibration

Horizontal axis: ring index I. Vertical axis: seasonal mean temperature T.

Estimate coefficients from paired observations.



Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Teaching model and units

Model: estimated temperature T_est = 6 + 4I.

I is dimensionless. The intercept is in °C and the slope in °C per index unit.

These coefficients belong only to this synthetic example.

R52 Yonenobu & Eckstein (2006)

Calibration and verification periods

Calibration: 2001–2004, used to estimate coefficients.

Calibration 2001–2004

Estimate coefficients

Verification 2005–2007

Evaluate the fixed equation

Both are short fictional periods for teaching.

Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Keeping verification independent

Season, variable and processing choices are part of model building.

Adjusting after viewing test results calls for a new independent evaluation.

Record the time split and search procedure.

R52 Yonenobu & Eckstein (2006)

Exercise A: predict unused years

Use T_est = 6 + 4I, then calculate error T_est − T.

Year	Index I	Observed T (°C)	Predicted T_est (°C)
2005	0.8	9.0	Calculate
2006	1.0	10.1	Calculate
2007	1.2	10.9	Calculate

8 minutes: calculate 4, compare 3, record 1.

Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Exercise A: estimates and errors

Predictions are 9.2, 10.0 and 10.8°C.

Year	Observed T (°C)	Predicted T_est (°C)	Error (°C)
2005	9.0	9.2	0.2
2006	10.1	10.0	-0.1
2007	10.9	10.8	-0.1

Agreement in this small example is not a general accuracy claim.

Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Correlation, bias and prediction error

Correlation measures shared variation.

Bias is systematic over- or underestimation.

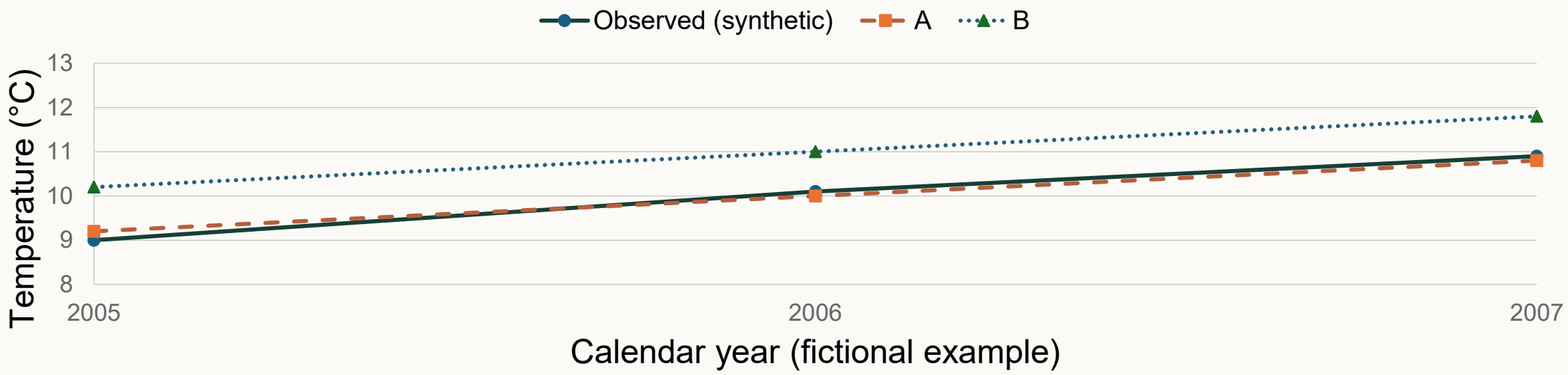
RMSE is the square root of mean squared error, in °C here.

R52 Yonenobu & Eckstein (2006)

Equal correlation with different bias

Model B adds 1°C to every estimate from model A.

The pattern is unchanged, but the errors grow.



Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Exercise B: compare prediction errors

Compare A, B and the constant prediction of 10°C.

Year	Observed (°C)	A (°C)	B (°C)	Constant (°C)
2005	9.0	9.2	10.2	10.0
2006	10.1	10.0	11.0	10.0
2007	10.9	10.8	11.8	10.0

6 minutes: calculate and discuss whether correlation is sufficient.

Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Exercise B: comparing RMSE

RMSE: A ≈ 0.14°C, B ≈ 1.01°C, constant ≈ 0.78°C.

Prediction	Correlation r	RMSE (°C)
A	0.996	0.14
B	0.996	1.01
Constant 10°C	Undefined	0.78

A three-year synthetic illustration of evaluation design.

Synthetic teaching data / schematic, not empirical results.

R52 Yonenobu & Eckstein (2006)

Changing responses through time

Divergence: growth–climate relationships may change through time.

Reconsider water stress, pollution, competition and physiology.

Use moving periods and independent evidence to assess stability.

R52 Yonenobu & Eckstein (2006)

Japanese case: Hinoki and early spring temperature

Yonenobu & Eckstein (2006): February–April temperature in central Japan.

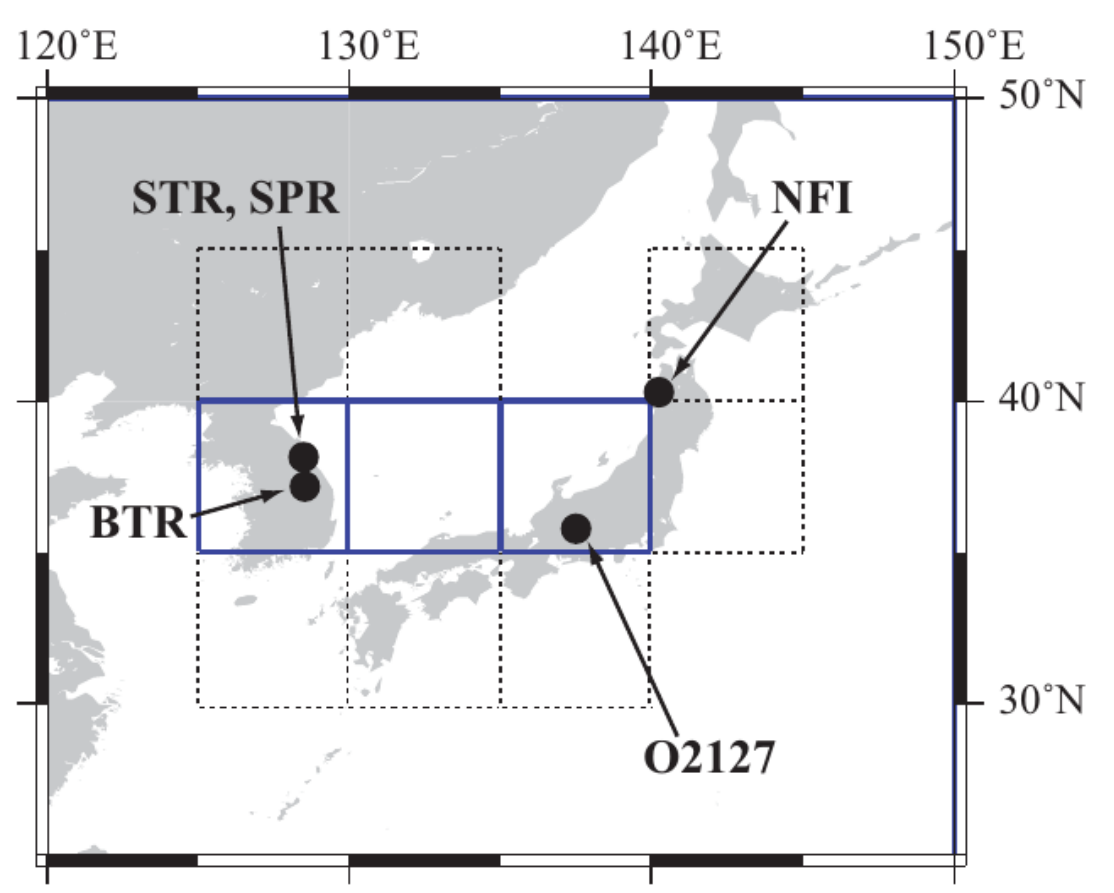
Stage	Period	Correlation r
Calibration	1900–1961	0.514
Verification	1962–2001	0.078

Reported in Table 1 of the paper, not reanalysed here.

R52 Yonenobu & Eckstein (2006)

Reconstructing Northeast Asian spring temperature

Distinguish sample sites from the climate target area.



Black dots: forest sites in the final calibration

Blue outline: March–May land-temperature target

Dotted outline: candidate climate grids

Five of seven candidate chronologies retained

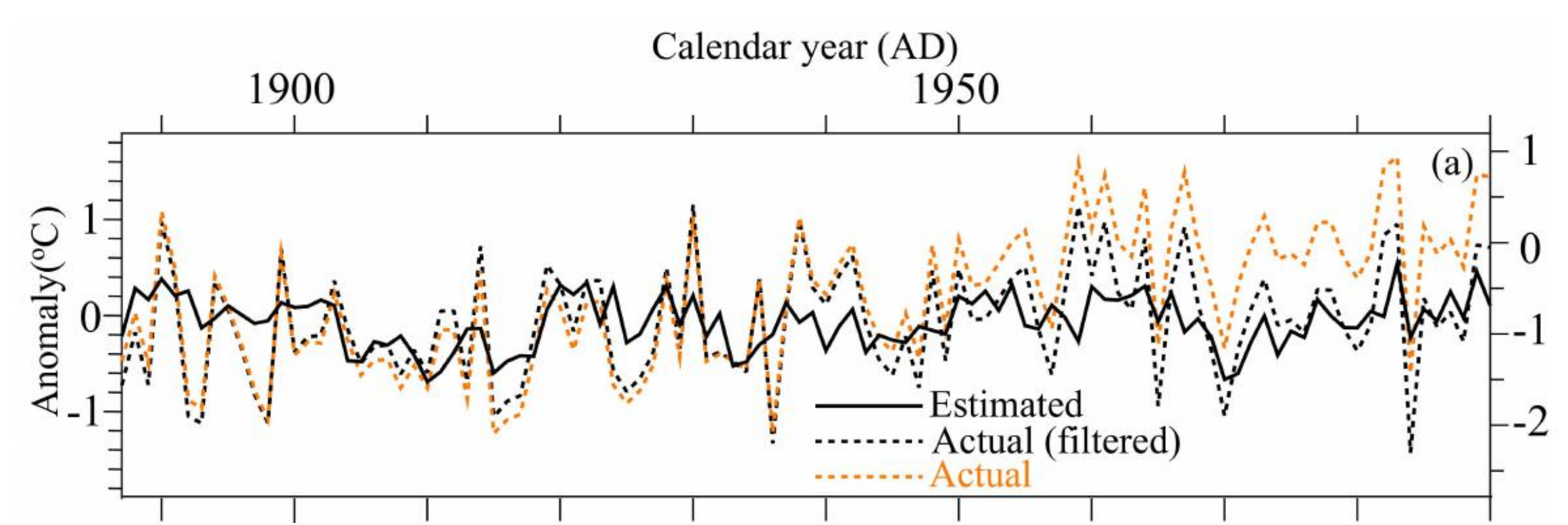
Reconstruction: 1784–1990 CE

Two chronologies originate at the same site.

Ohyama et al. (2013), Fig. 1 (cropped)

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Observed and estimated temperature: calibration



Black solid: estimate; black dotted: filtered observations (left axis)
Orange dotted: original observations (right axis)

1887–1990 calibration fit; this panel is not an independent test.

CE compares error with the verification-mean benchmark: 0.019 / −0.040.

Source: Ohyama et al. (2013), cropped from Fig. 3(a); Table 2

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Summer climate from stable isotopes

Cellulose oxygen isotopes reflect source water and leaf processes.

A central Japan study integrates oxygen and hydrogen isotopes.

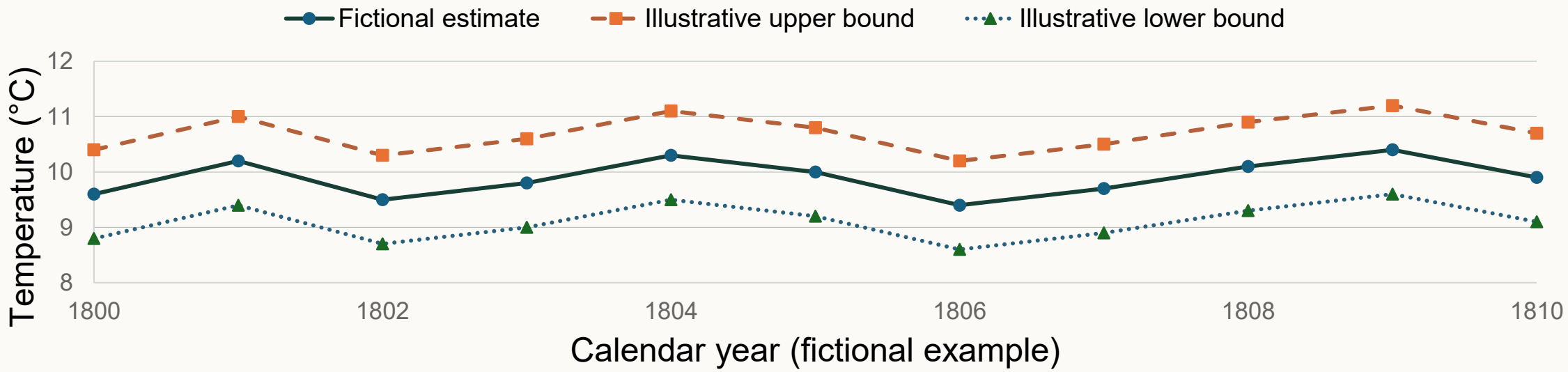
Check age and physiological effects in isotope records too.

R53 Nakatsuka et al. (2020)

Reconstruction curves and uncertainty

Show uncertainty and sample depth alongside the curve.

This fictional illustration is not a 95% interval.



Synthetic illustration. Bounds are chosen values, not statistical 95% intervals.

R52 Yonenobu & Eckstein (2006)

Spatial and seasonal scope

Do not equate one site's spring temperature with global annual temperature.

Test spatial similarities and differences against observations.

State region, elevation, season and proxy in the conclusion.

R52 Yonenobu & Eckstein (2006)

Extreme years and average relationships

A linear model may fail outside its calibration range.

Consider growth cessation, discontinuous rings and survival bias.

Distinguish identifying an extreme year from estimating its magnitude.

R52 Yonenobu & Eckstein (2006)

Reproducible analysis records

Retain data, units, missing values, processing and equations.

Record calibration and verification periods and selection decisions.

Label sources, periods and synthetic versus observed data.

R52 Yonenobu & Eckstein (2006)

Exit questions

Does high correlation imply little bias?

Why must season selection be separated from verification?

What information should accompany a reconstructed curve?

R52 Yonenobu & Eckstein (2006)

Next: a research plan or literature review

Lesson 8 integrates archaeological, ecological and climate evidence.

Separate evidence for dates, environment and causes.

Choose a research plan or a literature review for your question.

R52 Yonenobu & Eckstein (2006)